

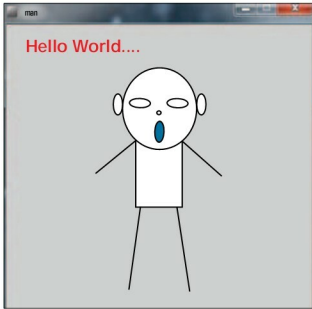


Figure 2: Static sketch output

```
ellipse(widthLen+25,heightLen-7,25,12);
ellipse(widthLen,heightLen+5,5,5);
fill(0, 102, 153);
ellipse(widthLen,heightLen+25,12,25);
fill(255,0,0);
textSize(32);
text("Hello World....", 10, 30);
line(widthLen+25, heightLen+125, widthLen+40, heightLen+225);
line(widthLen-25, heightLen+125, widthLen-40, heightLen+225);
```

This example shows how easy it is to get started with Processing.

Interactive sketch – ‘Hi dude!’: In this example, let’s change a static sketch into an interactive one. The following are the actions that the sketch will perform, based on user actions:

- If you move the mouse pointer, the sketch will move both hands in that direction.
- If you click the left mouse button, the left hand will be moved.
- If you click the right mouse button, the right hand will be moved.
- If you press the letter ‘A’ in the keyboard, the left leg will be moved.
- If the user presses the letter ‘D’ in the keyboard, the right leg will be moved.

```
void setup()
{
  // Initialize Screen size & draw sketch
}

void draw()
{
  if (mousePressed && (mouseButton==LEFT))
```

```
{
  // User Action to re-draw LEFT hand
  line(widthLen+25,heightLen+50,
widthLen+80+mouseX, heightLen+80+mouseY);
}
else if (mousePressed && (mouseButton==RIGHT))
{
  // User Action to re-draw RIGHT hand
}
else
{
  // default shapes
}
}

void mouseMoved() {
  // Action to take when mouse is moved without
  pressing
}
```

```
void keyPressed()
{
  if ((key == 'A') || (key == 'a'))
  {
    // re-draw left leg shape
  }
  if ((key == 'D') || (key == 'd'))
  {
    // re-draw right leg shape
  }
  else
  {
    //draw default shape
  }
}
```

In the above pseudo code *setup()*, *draw()*, *keypressed()* and *mouseMoved()* are built-in functions and are called internally, during execution. Similarly *key*, *mousePressed*, *mouseX*, *mouseY* and *mouseButton* are built-in system variables for handling user interactions.

An overview of libraries

The capabilities of the Processing application can be improved with the help of libraries. It supports the following third party libraries:

- DXF, PDF – to export 3D drawing data to the respective format.
- Serial – to interact with external hardware for sending/receiving data.
- Network – to send/receive data over the HTTP network.
- Audio/video – to capture inputs from camera, recorder or file and manipulate them.